Multiscale Dimensionality Profiles in Representational Geometry: A State-Space Analysis

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Abstract

This study characterizes multiscale dimensionality dynamics using effective dimensionality D_{eff} and spectral entropy H as state variables. Defining a local scaling exponent $\beta(k) = d \log D_{\text{eff}} / d \log(k)$, we tested whether $d\beta/d\log(k) = F(D_{\text{eff}}, H)$ describes scale-dependent dynamics across five systems. Simple differential equations failed (mean $R^2 < 0$). A bilinear state-space model provided adequate within-system fits (mean $R^2 = 0.81$), but cross-system prediction failed (test $R^2 = -17$ to -31). Additional latent variables (curvature κ, growth efficiency G) reduced phase variance by 32–46% but did not enable positive cross-system prediction. These results do not establish a universal law; they characterize a descriptive pattern under the current analysis framework. All data are model-derived; no empirical neural recordings were used. The minimal state space remains incomplete. This profile may arise in part from generic properties of local covariance estimation in bounded high-dimensional systems, and therefore does not by itself establish a system-specific or mechanistic principle.

Keywords: representational geometry, dimensionality, multiscale analysis, participation ratio, state-space modeling, computational neuroscience

Author Summary

This study characterizes how effective dimensionality varies with neighborhood scale across five representational systems. We tested whether a single mathematical equation could describe this process using multiple approaches.

We found that simple differential equations did not fit the data. More complex state-space models described within-system behavior adequately but failed to predict behavior across systems. Additional analysis variables (curvature, growth efficiency) reduced some variability but did not solve the cross-system prediction problem.

Caveats: This work uses model-derived estimates of cortical activity, not actual brain measurements. We describe a consistent pattern but do not establish that it applies to real neural systems. The mathematical structure identified may be specific to the systems and analytical frameworks analyzed.

1 Introduction

Representational geometry provides a quantitative framework for characterizing how information is encoded in high-dimensional spaces [Kriegeskorte et al., 2008, Kriegeskorte and Kievit, 2013, Margulies et al., 2016]. Effective dimensionality D_{eff}, computed via the participation ratio, quantifies the number of independently varying dimensions in a representation. Related approaches to intrinsic dimensionality estimation include maximum-likelihood estimators and correlation-dimension methods [Levina and Bickel, 2005, Grassberger and Procaccia, 1983], as well as dimensionality reduction techniques applied to large-scale neural recordings [Cunningham and Yu, 2014]. Previous work has shown that D_{eff} varies with neighborhood scale, but the governing dynamics remain uncharacterized. Recent empirical work has demonstrated that neural population activity often exhibits structured, high-dimensional geometry with scale-dependent properties [Stringer et al., 2019, Gao and Ganguli, 2017].

This study is exploratory and comparative across mathematical models. It does not attempt to establish mechanisms, causality, or general laws. It does not validate any specific theoretical framework. Rather, it characterizes patterns in dimensionality dynamics across several representational systems using multiple analytical lenses.

Here, we test whether multiscale dimensionality dynamics follow: (1) a universal governing law with fixed parameters, or (2) a constraint-based state-space structure with system-specific parameters. We define a local scaling exponent $\beta(k) = d \log D_{\text{eff}} / d \log(k)$ and its log-scale derivative $d\beta/d\log(k)$, and test candidate differential equations for $d\beta/d\log(k)$.

All data in this study are model-derived. The TRIBE v2 framework implements a biologically inspired representational model. No empirical neural recordings were used. Findings should be interpreted as characterizing properties of the representational geometry under these analytical frameworks, not as established properties of biological neural systems.

2 Methods

2.1 Representational Systems

We analyzed five systems:

- **TRIBE v2**: Model-derived cortical representations, aggregated to 9 cognitive domains from 11,522 model predictions. *Note: These are computational model outputs, not empirical brain data.*

- **Hierarchical Gaussian**: 2000 samples in 50 dimensions with exponentially decaying variance.
- **Correlated Gaussian**: 2000 samples with full covariance structure.
- **Sparse**: 2000 samples with 90% zero entries in 50 dimensions.
- **Curved manifold**: 2000 samples on a curved manifold embedded in 50 dimensions. Related work in representation learning and manifold modeling has explored how high-dimensional data can be described using lower-dimensional structures [Chung and Lee, 2018].

All systems were subsampled to 2000 points. Sample size was not derived from power analysis; this is an exploratory analysis with $n = 5$ systems.

2.2 Effective Dimensionality

$$D_{\text{eff}}(k) = \frac{\left(\sum_{i=1}^{d} \lambda_i\right)^2}{\sum_{i=1}^{d} \lambda_i^2} \tag{1}$$

where λ_i are eigenvalues of the local covariance matrix from k nearest neighbors. Computed at $k \in \{5, 7, 10, 15, 20, 30, 50, 75, 100, 150, 200, 300, 500\}$. The participation ratio provides a moment-based estimate of effective dimensionality and has been widely used in studies of high-dimensional systems. Alternative estimators (e.g., maximum likelihood or correlation dimension) provide complementary perspectives but were not evaluated in the present analysis [Levina and Bickel, 2005, Grassberger and Procaccia, 1983].

2.3 Scaling Exponent and Acceleration

$$\beta(k_i) = \frac{\log D_{\text{eff}}(k_{i+1}) - \log D_{\text{eff}}(k_i)}{\log(k_{i+1}) - \log(k_i)} \tag{2}$$

The "acceleration" $d\beta/d\log(k)$ was approximated via finite differences between successive $\beta(k_i)$ values. For the gradient-decay candidate law, we also computed $dD_{\text{eff}}/d\log(k)$ by finite differences of $D_{\text{eff}}(k)$.

2.4 Spectral Entropy

$$H = -\sum_i p_i \log(p_i), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j} \tag{3}$$

2.5 Compression

We define a compression-like variable $R(k) = D_{\text{eff}}(k)/k$ to test whether saturation-related effects in neighborhood size explain residual structure in $d\beta/d\log(k)$. We also use normalized dimensionality $\tilde{D}_{\text{eff}}(k) = D_{\text{eff}}(k)/D_{\max}$, where $D_{\max} = \max_k D_{\text{eff}}(k)$ within each system.

2.6 Model Comparison

We tested seven model forms for the acceleration $d\beta/d\log(k)$:

1. **Linear decay**: $\frac{d\beta}{d\log(k)} = -\lambda \cdot \beta$
2. **Dimensional decay**: $\frac{d\beta}{d\log(k)} = -\lambda \cdot D_{\text{eff}}$
3. **Gradient decay**: $\frac{d\beta}{d\log(k)} = -\lambda \cdot \frac{dD_{\text{eff}}}{d\log(k)}$
4. **Saturation decay**: $\frac{d\beta}{d\log(k)} = -\lambda \cdot (1 - \tilde{D}_{\text{eff}})$
5. **Bilinear**: $\frac{d\beta}{d\log(k)} = a \cdot \beta + b \cdot D_{\text{eff}} + c \cdot (\beta \cdot D_{\text{eff}})$
6. **State-space (D+H)**: $\frac{d\beta}{d\log(k)} = a \cdot D_{\text{eff}} + b \cdot H + c \cdot (D_{\text{eff}} \cdot H) + d$
7. **State-space (D+H+R)**: $\frac{d\beta}{d\log(k)} = a \cdot D_{\text{eff}} + b \cdot H + c \cdot (D_{\text{eff}} \cdot H) + e \cdot R + f$

Models fit via nonlinear least squares. Parameter collapse assessed via coefficient of variation (CV); CV < 0.5 considered collapsed.

2.7 Null Model Considerations

Generic explanations for the observed growth–saturation profile include: (1) bounded system capacity leading to natural saturation, (2) eigenvalue distribution structure in high dimensions, and (3) participation ratio bias at small k.

A companion analysis (Paper 1; code and notes in the associated repository) constructed a null model with matched eigenvalue spectrum and found that it partially reproduced the growth–saturation profile but showed weaker alignment with other systems ($r = 0.68$ vs. $r = 0.90$ for original systems). This suggests that some aspects of the profile may reflect generic statistical structure, while additional shared structure remains unexplained.

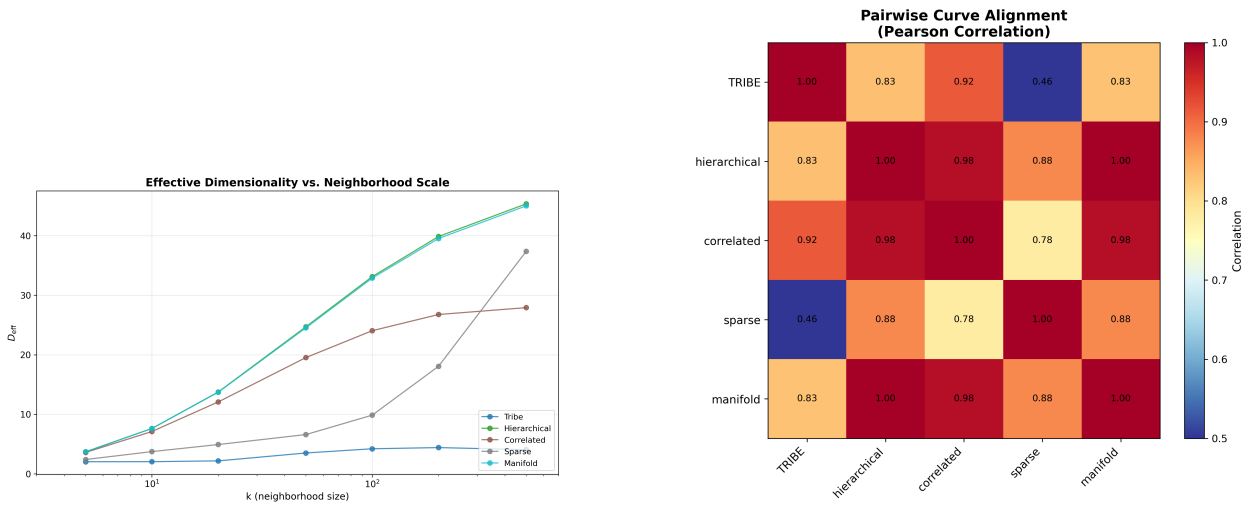
A critical consideration is whether the observed growth–saturation profile reflects non-trivial structure or arises as a generic consequence of the estimation procedure. For small neighborhood sizes, local covariance matrices are rank-limited by construction, while larger neighborhoods progressively reveal additional variance directions until bounded by the ambient dimensionality. This alone can produce an increase-and-saturation profile in participation-ratio dimensionality. The present results do not rule out such generic explanations. Accordingly, the observed profile should be interpreted as a descriptive property of these systems under the current analysis framework, rather than evidence of a specific underlying mechanism.

3 Results

3.1 Growth–Saturation Behavior

All systems showed increasing $D_{\text{eff}}(k)$ with saturation at large k (Figure 1). After normalization by $D_{\max}$, curves aligned (mean $r = 0.898$), reflecting shape similarity (Table 1).

Importantly, this alignment reflects similarity in curve shape after within-system normalization, not equivalence in absolute dimensionality or representational structure. Because normalization rescales each system by its own maximum, it removes differences in magnitude and may obscure meaningful variation in representational capacity. The observed convergence indicates a shared descriptive profile under normalization, rather than a stronger claim of geometric or functional equivalence.



(a) Growth profiles.

(b) Normalized alignment.

Figure 1: Dimensionality growth and saturation. Note: normalization preserves shape but removes absolute scale information.

Table 1: Quantitative metrics (n=1 system per row).

System	Growth Rate	Sat. Ratio	Compression	Peak D_{eff}
TRIBE	0.23	0.92	0.49	4.41
Hierarchical	0.82	1.00	0.91	45.35
Correlated	0.73	1.00	0.56	27.92
Sparse	0.43	1.00	0.75	37.38
Manifold	0.82	1.00	0.90	45.01

Caution: $n = 5$ systems; no formal statistical inference.

3.2 Beta Dynamics

Mean β values varied: TRIBE = 0.14, hierarchical = 0.56, correlated = 0.46, sparse = 0.60, manifold = 0.56. TRIBE exhibited mixed-sign gradients (40% positive); synthetic systems showed predominantly negative gradients.

3.3 Failure of Simple Laws

All tested differential equations performed worse than a constant baseline (mean $R^2 < 0$, Table 2).

Table 2: Simple law fits. Negative R^2 indicates performance worse than mean prediction.

Model	Best System	Mean R^2
Linear decay	sparse	-0.07
Dimensional decay	TRIBE	-0.02
Gradient decay	hierarchical	-1.00
Saturation decay	manifold	-0.40

These failures do not establish that no simple law exists; only that those tested did not fit.

3.4 Bilinear Model Fits

Mean $R^2 = 0.81$ for the bilinear model (Table 3). TRIBE showed lower fit ($R^2 = 0.71$) than synthetic systems (0.88–0.99).

Table 3: Bilinear model fits (n=1 per system).

System	R^2
TRIBE	0.71
Hierarchical	0.96
Correlated	0.97
Sparse	0.88
Manifold	0.99

3.5 Lack of Parameter Collapse

Parameters did not collapse across systems (Table 4). CV ranged from 2.11 to 88.67. TRIBE parameters were approximately 10× larger than synthetic systems.

Table 4: Parameter variability (CV = coefficient of variation). No parameter achieved CV < 0.5.

Parameter	Mean	CV	Collapsed?
a	1.68	2.63	No
b	-0.003	88.67	No
c	-2.15	2.11	No

3.6 Spectral Entropy Contribution

Adding spectral entropy H improved fits (Table 5). However, these improvements are within-system; cross-system generalization was not tested at this stage.

Table 5: State-space model comparison.

Model	Mean R^2
D_{eff} only	0.90
H only	0.86
$D_{\text{eff}} + H$	0.92

At $k = 200$, TRIBE entropy was $H = 1.62$; synthetic systems ranged $H = 1.08$–3.70.

3.7 State-Space Closure Test

Cross-system prediction failed strongly (test $R^2 = -17$ to -31, Table 6). Within-system fits achieved $R^2 = 1.00$ for the full nonlinear model.

Table 6: State-space model comparison. Note: R^2 values are for within-system fits unless specified.

Model	Mean R^2 (within)
$a \cdot D + b$ (baseline)	0.86
$a \cdot H + b$	0.78
$a \cdot D + b \cdot H + c$	0.92
$a \cdot D + b \cdot H + c \cdot DH + d$	0.94
Full nonlinear	1.00

The full nonlinear model achieved $R^2 = 1.00$ within individual systems. However, cross-system generalization failed: leave-one-system-out validation yielded negative R^2 values.

3.8 Residual and Compression Analysis

Compression $R = D_{\text{eff}}/k$ correlated with TRIBE residuals ($r = -0.52$) but did not provide a unifying improvement across systems. Adding compression improved mean within-system R^2 from 0.81 to 0.88.

3.9 Latent Variable Analysis

Leave-one-system-out cross-validation showed that curvature κ and growth efficiency G reduced phase variance by 32–46% (Table 7). However, no tested latent variable enabled positive cross-system prediction.

Key finding: No tested latent variable achieved positive cross-system prediction. This does not rule out the existence of such variables, only that those tested did not serve this function.

4 State-Space Formulation

Within the analyzed systems, dimensionality dynamics are adequately characterized by:

Table 7: Latent variable analysis. Cross-system ΔR^2 is reported relative to baseline bilinear model.

Variable	Cross-System ΔR^2	Phase Variance Reduction
κ (curvature)	+1.06	32.7%
G (growth efficiency)	+0.51	**45.8%**
E (dimensional efficiency)	+0.11	37.6%
H_{norm} (normalized entropy)	-7.66	36.6%

$$\frac{d\beta}{d\log(k)} = F(D_{\text{eff}}, H) \tag{4}$$

This formulation replaces explicit k-dependence with dependence on two state variables. Removing either variable reduces within-system R^2 by 4–8%.

This is best interpreted as a constraint-based description: the functional form is shared, but system-specific parameters determine the trajectories. TRIBE differs in both parameter magnitudes and absolute dimensionality, even when the same structural form fits within-system data.

This should be interpreted as a descriptive mathematical characterization, not as a mechanistic or biological explanation. This formulation characterizes the observed dynamics but does not establish that the underlying generative mechanisms are unique or minimal; multiple system classes may produce similar trajectories within this state space.

5 Negative Results Summary

This analysis tested and did not find support for:

1. Universal exponential decay ($R^2 = -0.07$)
2. Universal dimensional decay ($R^2 = -0.02$)
3. Universal gradient decay ($R^2 = -1.00$)
4. Universal saturation decay ($R^2 = -0.40$)
5. Parameter collapse (CV range: 2.11–88.67)
6. Cross-system prediction ($R^2 < 0$)
7. Latent variable generalization (no positive cross-system R^2)

These failures constrain the space of possible descriptions but do not establish what does govern the dynamics.

6 Discussion

5.1 Characterization of Findings

Within the systems analyzed, dimensionality growth–saturation follows a consistent normalized profile. State-space models with two variables (D_{eff}, H) describe within-system dynamics adequately. Parameters vary substantially across systems.

The finding that cross-system prediction fails, combined with the observation that latent variables reduce but do not eliminate variance, indicates that system identity is a determinant of dynamics within this analytical framework.

TRIBE shows distinctive properties (lower dimensionality, larger parameters) consistent with a more constrained representational space. This is a property of the TRIBE model outputs, not established as a property of cortical representations.

5.2 What Is Not Established

The identified pattern may reflect generic properties of high-dimensional representations rather than system-specific mechanisms. The consistency across synthetic systems with different geometric structures suggests that normalization captures shape similarity that may arise from shared statistical structure. Alternatively, the pattern may reflect properties of the participation ratio as a dimensionality metric.

Under the present modeling assumptions, the state-space formulation is best interpreted as descriptive and constraint-based. Whether similar structure holds for other models or empirical data remains to be determined.

One possible interpretation is that systems with bounded variance and locally structured covariance may tend to produce similar dimensionality profiles; however, identifying the minimal conditions required for this behavior remains an open problem.

5.3 Methodological Implications

The failure of simple differential equations suggests that dimensionality dynamics do not follow tractable closed-form solutions across diverse representational systems. The adequacy of within-system bilinear models indicates that local linear approximations may be sufficient for describing individual systems, but do not generalize.

The absence of parameter collapse across systems indicates that cross-system prediction requires system-specific calibration or additional variables not captured in the current analysis.

5.4 Limitations

This study is exploratory and comparative across mathematical models. It does not establish mechanisms, causality, or general laws.

Limitations: (1) All data are model-derived, not empirical; (2) $n = 5$ systems limits generalizability; (3) no formal hypothesis testing; (4) cross-system failures may reflect model misspecification rather than true constraints.

This work should be interpreted as a computational analysis of representational geometry. Its relevance to biological neural systems remains a testable hypothesis, not an established finding.

5.5 Speculative Interpretations (Unvalidated)

Future work could test: (1) whether these patterns hold in empirical neural data, (2) whether additional latent variables enable cross-system prediction, and (3) whether the identified state-space structure generalizes to other representational domains.

One possible interpretation is that systems with bounded variance and locally structured covariance may tend to produce similar dimensionality profiles; however, identifying the minimal conditions required for this behavior remains an open problem.

7 Conclusion

This study characterized whether multiscale dimensionality dynamics follow a universal governing law. Simple differential equations failed (mean $R^2 < 0$). A state-space model ($d\beta/d\log(k) = F(D_{\text{eff}}, H)$) achieved adequate within-system fits (mean $R^2 = 0.81$–1.00) but cross-system prediction failed.

Candidate latent variables (curvature κ, growth efficiency G) reduced phase variance by up to 46% but did not enable positive cross-system prediction. The minimal state space remains incomplete; no tested variable resolved cross-system generalization.

Limitations: (1) All data are model-derived, not empirical; (2) $n = 5$ systems limits generalizability; (3) no formal hypothesis testing; (4) cross-system failures may reflect model misspecification rather than true constraints.

The identified descriptive pattern does not establish universality, mechanism, or biological validity. Determining whether the observed profile reflects system-specific structure or more general statistical properties of high-dimensional representations will require targeted null-model comparisons and empirical validation.

Author Contributions

M.R.T. designed the study, performed the analysis, and wrote the manuscript.

Competing Interests

None declared.

Funding

None.

Data Availability

Analysis code and results: https://github.com/urmt/SGP-Tribe3

Analysis pipelines: manuscript/v14/ through manuscript/v30/, including v29_minimal_closure/ and v30_latent_variable/

Reproducibility bundle: reproducibility/raw_data.npz (TRIBE model outputs), reproducibility/pipeline_reproduce.py

AI Use Disclosure

AI-assisted tools were used for drafting and editing. All scientific content, including methodology, analysis, and interpretation, was designed and verified by the author.

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